

Assignment: AI-assisted Code Reviews

Overview: This assignment tests your individual capability of applying the skills acquired during this course to a software engineering context. The assignment consists of a case description paired with data sets and several tasks.

General notes: When using material outside of the official course material (lectures, labs, assignments), state the source and critically reflect on its usage. In case of questions, please contact Julian Frattini (julian.frattini@chalmers.se).

1. Case Description

At a software development company, some engineers have started to use artificial intelligence (AI) tools to support their code review (CR) activity. The company wants to understand whether AI-supported CRs have an effect on the (1) extent and (2) quality of their CRs.

The company does have a lot of historical data on the use of AI in CRs: They have collected records of the resulting CRs' extent (i.e., their length in number of characters), their quality (assessed by experts on a 1-5 scale), but analyzing the impact is not trivial. On the one hand, some CRs apply to code on more complex issues, others on simpler issues, which might distort the effect. On the other hand, the company has no control over who uses AI for CR, but younger engineers are much more open to using AI tools. Older engineers are adamant that the defining factor for concise and high-quality CRs is skill, and skill only comes with time.

The aversion also manifests in how these CRs are perceived: Previously, engineers assumed that whether a CR was considered relevant for making a decision (e.g., whether to approve a pull request or not) depended only on how extensive and how high quality it was. But recently, a clear aversion against AI-assisted CRs is notable, as AI-written CRs are categorically discarded more often.

2. Tasks

Deliver a report containing solutions to the following tasks:

1. Formalize your causal assumptions about the phenomenon under investigation based on the case description above. Propose at least two **directed acyclic graphs** (DAGs) and discuss their viability. Every DAG should contain *all* relevant outcome variables.
2. Visualize important aspects of the data with **descriptive statistics**. Normalize/standardize data where necessary and appropriate.
3. Derive **statistical models** in mathematical notation from the causal models. For this, choose one of the following two options:
 - (a) Select one of the presented DAGs and derive one statistical model *per outcome variable*.
 - (b) Select one outcome variable that is present in all presented DAGs and derive one statistical model for this particular outcome variable *from each DAG*.¹
4. Present and discuss **sensible priors** for all coefficients in your statistical model (including prior predictive checks).
5. **Assess the training** of the statistical models. If you chose variant (b), **compare the two models**.
6. Draw a **conclusion about the effect** of interest based on your analysis and model comparison.

¹This option is only viable if the statistical models differ for different DAGs.

3. Available Data

You find two data sets next to the assignment description. These data set contain the following variables.

3.1. Reviewers

Variable	Description	Values
reviewer.id	Identifier of the reviewer	1-200
age	Age of the reviewer in number of years	[20, 65]
skill	Skill of a reviewer measured on a continuous scale	[0.0, 5.0]
used.cr.technology	Whether the reviewer uses AI-support during code review (1) or not (0)	0/1

3.2. Reviews

Variable	Description	Values
reviewer.id	Identifier of the reviewer that conducted the review	1-200
review.id	Identifier of the code review	1-2000
complexity	Relative complexity of the code change to be reviewed	$\mathcal{N}(0, 1)$
result	Whether the code review contributed to making a decision on the pull request (1) or not (0)	0/1
extent	Number of characters in the code review	$\mathbb{N}$
quality	Subjective quality of the code review as assessed by experts (where 1 is bad and 5 good quality)	1-5

4. Notes

This section contains some additional notes that may guide your submission:

1. Do not expect to fit *everything* into your report (be aware of the submission instructions including a page limitation). Focus on relevant and important pieces of information.
2. 2-4 descriptive statistics of the data suffice.
3. You do **not** need to apply multi-level modeling in tasks 3-6. You can safely ignore multi-level models in this assignment.